Quiz 4 - Computational Physics I

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Date: Thursday 29 May 2025 **Duration:** 45 minutes

Credits: 20 points (5 questions) **Type of evaluation:** LAB

Provide concise answers to the following items:

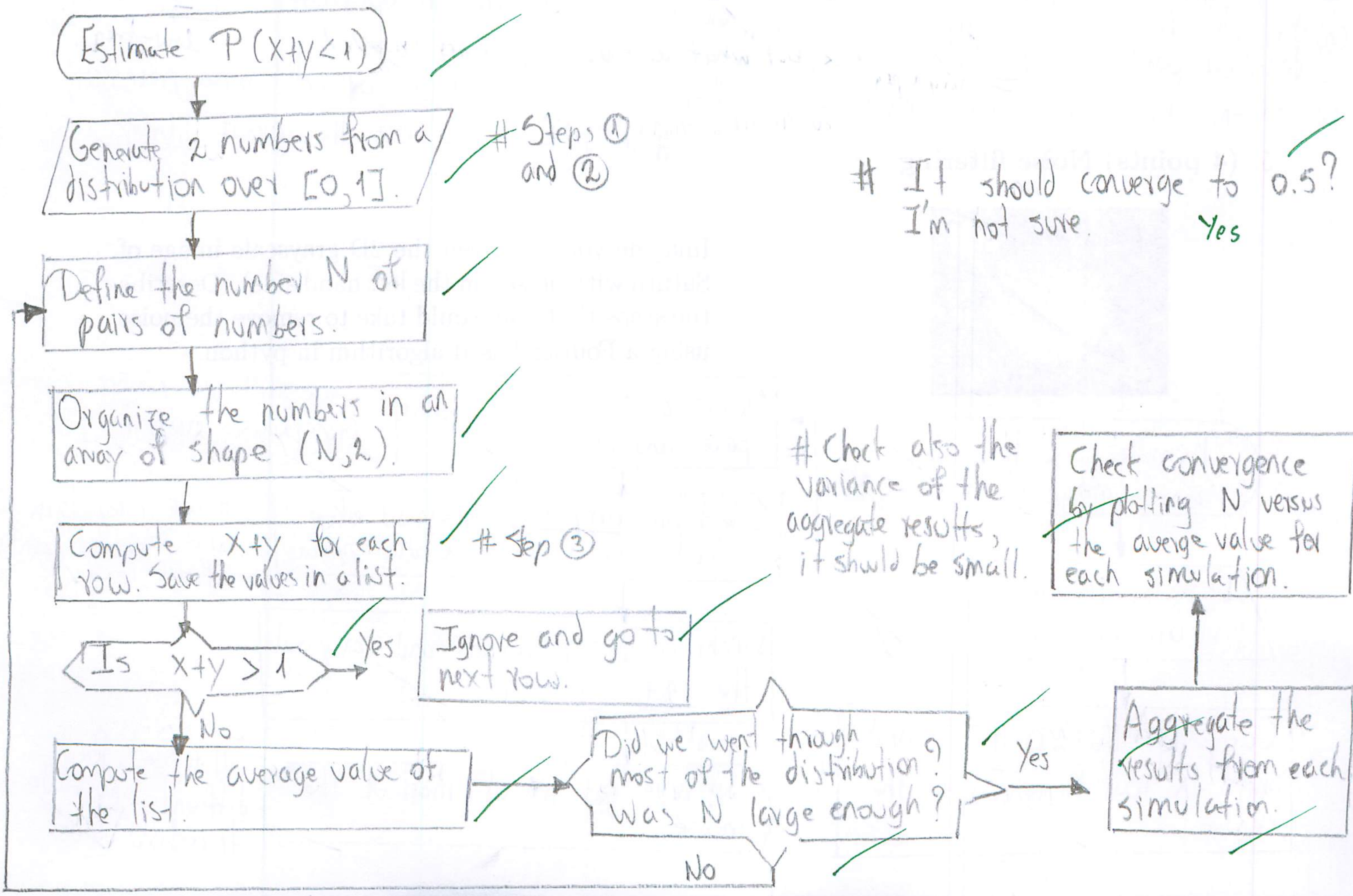
1. **(4 points) Monte Carlo methods**
   (a) What do Monte Carlo methods consist of?
   (b) What are the basic steps that you would follow to carry out a Monte Carlo simulation?

They're a type of computational algorithms that rely on repeated random sampling to converge to a numerical result, which is a solution to a deterministic problem very hard (or impossible) to solve analytically.

The basic steps are the following:

① Identify the domain of the input values.
② Generate random input values ~~over~~ from a distribution defined over the domain.
③ Carry out the deterministic computations by running a lot of simulations, enough so that it went through most of the infinite possible combinations.
④ Aggregate results and check convergence (i.e. variance between each simulation result).

2. **(4 points) Monte Carlo simulation**
   Imagine you are asked to estimate the probability that the sum of two random numbers (x and y) drawn uniformly from the interval $[0, 1]$ is less than 1 (i.e., $x + y < 1$). Describe the python workflow that you would use to estimate the probability $\mathcal{P}(x + y < 1)$. What value should it converge to?



3. **(4 points) Fourier filters**
   (a) Explain the difference between a low-pass filter and a high-pass filter in Fourier analysis.
   (b) List the main steps you would take to detect edges in images using Fourier analysis in python.

A low-pass filter over a signal selects just small frequencies in Fourier space, thus enhancing large-scale structure in real space after carrying out the I.F.T.

A high-pass filter over a signal selects just high frequencies in Fourier space, thus enhancing the details of the signal in real space after carrying out the I.F.T.

Steps to detect edges in images:
① Select one of the channels of the image.
② Carry out the 2D F.F.T
③ Shift the low frequencies to the center.
④ Define a mask so that everything far from the center is 1, and near the center zero. → high-pass filter
⑤ ~~Carry out the 2D~~ Multiply the mask times the 2D F.F.T of the image.
⑥ Unshift the frequencies and carry out the 2D I.F.T.
Do this for every channel of the image.

4. **(4 points) Wavelet transforms**
   (a) What is the difference between Fourier transforms and wavelet transforms?
   (b) Explain the process of wavelet decomposition for a 1D signal.

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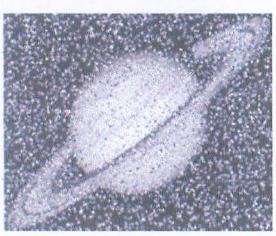
Fourier transforms can just be used when analyzing stationary signals (time or space). To analyze non-stationary signals (they change in space and time) we use wavelet transforms, which use wavelets ~~as~~ as basis functions that can be stretched, shrunken and translated to analyze the signal.

For a 1D signal:
① Choose our mother wavelet (haar, etc.)
② Multiply the wavelet times the signal, we get approximation coefficients (CA) and detailed coefficients (CD)
low-freq
high freq
OK, but what do these coefficients encode?
③ Do this until there's no clear difference between CA and CD, or until when necessary.
in python we use the pywt library.

5. **(4 points) Noise filtering**



Imagine you are given the 2D grayscale image of Saturn with noise (on the left hand side). Describe the steps that you would take to remove the noise using a Fourier-based algorithm in python.

Remove noise from Saturn image
↓
I/O the image, we have just one channel.
↓
Carry out the 2D F.F.T and shift the low frequencies to the centre.
→
Select a 1D slice of Fourier space and plot. # we still see noise, it populates everything.
↓
Select an amplitude threshold value, so that most of the noise dissappears. # we may lose some information/parts of the image.
↓
Unshift the frequencies and carry out the 2D I.F.T.
↓
We have get rid of most of the noise. # repeat for different amplitude threshold values.